

# Griffon McMahon

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## EDUCATION

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### University of Pennsylvania

*Master of Science in Engineering, Robotics*

Philadelphia, PA

May 2022

### Purdue University

*Bachelor of Science in Mechanical Engineering, Minor in Classical Studies*

West Lafayette, IN

May 2020

## EXPERIENCE

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### Graduate Research Fellow

*ScalAR Lab, University of Pennsylvania*

August 2022–May 2025

Philadelphia, PA

- Examined the effects of inserting a paucity of agents into a swarm with the goal of influencing the behavior of the whole
- Quantified uncertainty in unpredictable systems with conformal prediction used in the realm of target tracking
- Tracked agents with unknown dynamics in the presence of occlusions and faulty models

### Naval Research Enterprise Internship Program (NREIP)

*United States Naval Research Laboratory*

July 2024–September 2024

Washington, DC

- Analyzed many agents swarming modeled through delay differential equations using mean field analysis
- Employed bifurcation theory to determine the conditions in which a swarm would switch from one behavior to another

### Graudate Research Assistant

*Kod\*lab, University of Pennsylvania*

January 2021–May 2022

Philadelphia, PA

- Developed gaits for Jerboa, an underactuated and tailed bipedal robot, to take advantage of internal degrees of freedom for improved locomotion
- Analyzed dynamics behind hybrid, non-linear systems—especially legged robotics—using templates and anchors
- Implemented control algorithms in both simulation and hardware using event-based guards and feedback control

### Undergraduate Research Assistant

*Jain Research Lab, Purdue University*

June–December 2018, August 2019–May 2020

West Lafayette, IN

- Maintained and developed code using partially-observable Markov decision processes and Bayesian statistics to implement machine learning
- Authored conference paper applying clustering algorithms to human behavior
- Conducted human subject research to sense trust in automated systems using real-time sensors
- Crafted experiment design for examining human-machine interactions in the case of autonomous vehicles using a driving simulator

### Teaching Assistant

*MEAM 620: Advanced Robotics, University of Pennsylvania*

Fall 2021, Springs 2022, 2024–25

Springs 2022, 2024–25

- Instructed students on Kalman filters (for simulated IMU and camera data), PID control, and simulation to hardware tuning and deployment
- Guided students in deploying code on quadrotors (Crazyflie 2.1) for the purposes of control, trajectory planning, and path finding
- Maintained drone hardware throughout crashes stemming from student-made code
- Led team of TAs to ensure timeliness and productivity

*MEAM 520: Introduction to Robotics, University of Pennsylvania*

Fall 2021

- Assisted students with manipulator control of a Franka Emika Panda Arm in simulation and hardware for a pick and place task

### Awards and Honors

*First Place, 10th F1TENTH Autonomous Grand Prix at ICRA*

2022

- Won international competition undefeated using model predictive control (MPC), NVIDIA Jetson, and hardware-friendly trajectory planning

*Master's Award for Teaching*

2022

## TECHNICAL SKILLS

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**Languages:** MATLAB, Python, C/C++, Javascript, HTML/CSS

**Miscellaneous Software:** ROS/ROS2, Gazebo, Linux Computing, SolidWorks, LabVIEW, Git, Wolfram Mathematica

**Other:** Kinematics, Reinforcement Learning, MPC, Hybrid/Nonlinear Systems, Neural Networks, Sensor Integration/Filtering, Image Processing

## PUBLICATIONS

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- **G. McMahan**, K. Akash, T. Reid, and N. Jain, “On Modeling Human Trust in Automation: Identifying distinct dynamics through clustering of Markovian models.” *IFAC-PapersOnLine*, 2020
- K. Akash, **G. McMahan**, T. Reid, and N. Jain, “Human Trust-based Feedback Control: Dynamically varying automation transparency to optimize human-machine interactions.” *IEEE Control Systems Magazine*, 2020